2D CFD Analysis of Lift in a Formula One Racing Car

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Abstract: This study presents a two-dimensional *computational fluid dynamics (CFD)* analysis of a simplified Formula One racing car geometry, with the objective of investigating the aerodynamic flow features responsible for downforce generation, using ANSYS Fluent. The numerical simulations focus on key physical quantities such as pressure distribution, velocity field, and lift coefficient in order to characterize the global flow behavior around the vehicle. Particular attention is given to the extreme scenario in which aerodynamic forces could, in principle, sustain the car against gravity when driving on the ceiling, providing a conceptual assessment of whether sufficient negative lift may be generated under high-speed conditions.

A systematic mesh refinement and convergence study is conducted to assess the numerical consistency and reliability of the adopted approach. Although the inherent limitations of a two-dimensional model and the absence of three-dimensional effects are acknowledged, the study aims to provide qualitative insights into the aerodynamic mechanisms at play. The simulations are performed under steady-state conditions, using boundary conditions representative of a wind tunnel setup, and a turbulence modeling strategy guided by near-wall resolution requirements. The final converged solution yields a negative lift coefficient of approximately $C_L \approx -1.20$, dominated by pressure forces and primarily generated by strong ground-effect-induced pressure gradients beneath the car. Despite the simplified two-dimensional framework, the results indicate that, under extreme high-speed conditions, the predicted aerodynamic downforce is sufficient to overcome gravitational forces, supporting the conceptual feasibility of inverted driving in an idealized scenario.

I. Introduction

A. *Context and goals of the study*

Aerodynamic downforce is a key factor in racing car performance, directly affecting vehicle stability and tire grip. Computational fluid dynamics is widely used in motorsport engineering to analyze flow structures and aerodynamic forces during early design stages and for parametric studies. Simplified numerical models, while limited, remain useful for isolating dominant flow phenomena and assessing numerical methodologies.

In this study, a two-dimensional CFD approach is adopted to analyze the flow around a simplified Formula One racing car profile, using ANSYS Fluent. The objective is to investigate lift generation mechanisms and to evaluate the robustness of the numerical solution through mesh convergence, within the constraints imposed by the geometric and physical simplifications [1].

The main objective of this work is to analyze the aerodynamic forces acting on a Formula One racing car

profile and to assess the magnitude of the generated downforce under steady flow conditions. A secondary objective is to explore an extreme, idealized case in which the aerodynamic lift could counteract gravity, serving as a reference for evaluating pressure and velocity distributions.

Previous studies on racing car aerodynamics have demonstrated the importance of pressure gradients and flow acceleration in downforce generation (see, for example: [2], [3], [4]). While high-fidelity analyses typically rely on three-dimensional simulations and experimental testing, two-dimensional CFD models are commonly used for preliminary investigations and educational purposes. The present study aims to provide a quantitative assessment of these aerodynamic mechanisms and to verify the numerical consistency of the adopted approach.

B. Methodology

A two-dimensional Formula One racing car geometry is placed in a computational domain representative of an idealized wind tunnel test section [5]. Steady, incompressible turbulent flow conditions are assumed, consistently with the high Reynolds number regime typical of Formula One aerodynamics and with the focus on time-averaged aerodynamic loads.

A uniform velocity inlet is prescribed at the upstream boundary to reproduce a controlled free-stream condition, as commonly adopted in wind tunnel experiments, where the inflow velocity is known and imposed. At the downstream boundary, a pressure outlet condition is applied, fixing the static pressure to the ambient value. This choice allows the flow to naturally adjust its exit velocity and direction, minimizing artificial constraints on the wake development and ensuring a physically consistent pressure recovery far from the car.

No-slip boundary conditions are imposed on the car surface, enforcing zero relative velocity between the fluid and the solid walls, as required to correctly model boundary-layer formation, viscous shear stresses, and pressure forces contributing to lift and drag. At the upper boundary of the computational domain, a symmetry condition is applied. This choice reflects the assumption that the domain is sufficiently large for vertical velocity gradients and shear stresses to be negligible at that location, thereby approximating an unconfined flow while reducing computational cost and avoiding spurious wall effects.

The ceiling, acting as the moving reference surface along which the car travels in the present case study, is modeled as a moving wall with velocity equal to the inlet flow velocity. This modeling choice is essential to correctly reproduce the relative motion between the car and the ground, eliminating the artificial boundary layer that would otherwise develop on a stationary wall. As a result, the near-ground flow more accurately represents real on-track conditions and allows a physically meaningful investigation of ground-effect-induced pressure gradients beneath the car.

The numerical solution is obtained using a single turbulence modeling approach (Section B), with near-wall mesh refinement guided by y^+ requirements to ensure adequate resolution of the boundary layer. A mesh convergence study is carried out by progressively refining the grid, with increased resolution in regions characterized by strong velocity and pressure gradients. Convergence is assessed through both residual reduction and stabilization of global aerodynamic coefficients, ensuring numerical robustness and physical consistency of the obtained solution.

II. Geometric modeling and hypotheses

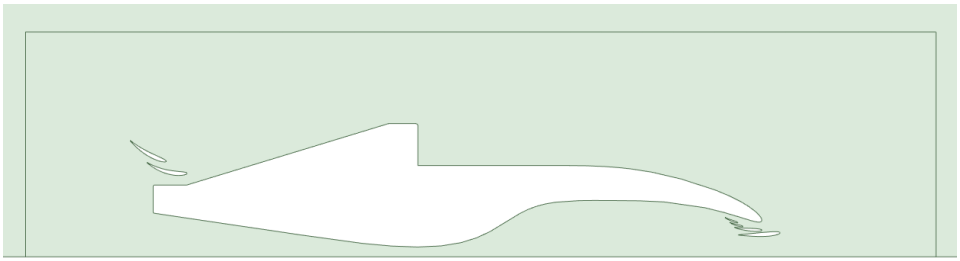
A. *Geometry*

Figure 1 shows the 2D model used for CFD analysis. A common practice in these cases is to consider a large portion of the fluid [6]. In front of the car, the fluid length is considered twice the car length, whereas on the back it is five times and once on top (in this case the length of the body without wings is considered as car length). The distance from the ground plays an important role in the simulation, since it is the main source of lift generated by the ground effect. In this case, we set the lowest point of the car to be at a distance of 70 mm.

An important element to highlight is the enclosure region close to the car. This separation has been added since it can be useful to refine the mesh to better study the fluid behavior around the car.



(a)



(b)

Figure 1: 2D design of the fluid

To obtain the fluid region, a CAD of the car has been created [2].

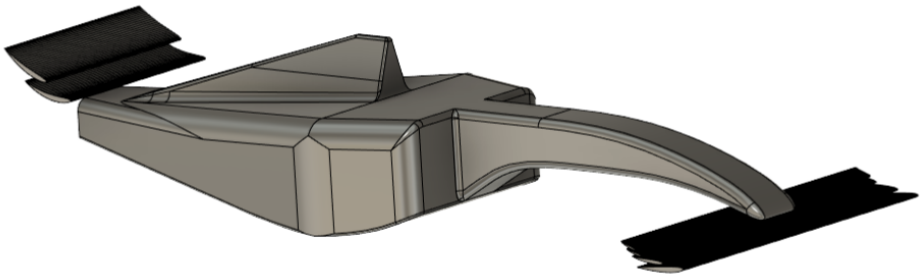


Figure 2: 3D CAD of the car

Since the interest is in the 2D profile of the car, the focus of the model is more centered on the front and rear wing and the body, especially the floor area, where a modeling of the ground effect is needed for generating lift. Therefore, the model does not include wheels with their suspensions as well as wing supports. Nowadays, Formula One cars have the halo designed for safety reasons, and since it does not significantly impact the lift study, it has been neglected. An important simplification of the

model concerns the front wing: since the 2D profile has been taken in the middle of the car, as it can be seen from figure 1b, the front wing is not connected with the nose, a small gap was left purposely. If the model was completely faithful to the real car, in the central profile, the nose would cover the front wing elements from the free stream completely eliminating the lift contribution.
The Formula One world is very secret around car characteristics, therefore the model was conceived with reference to images.

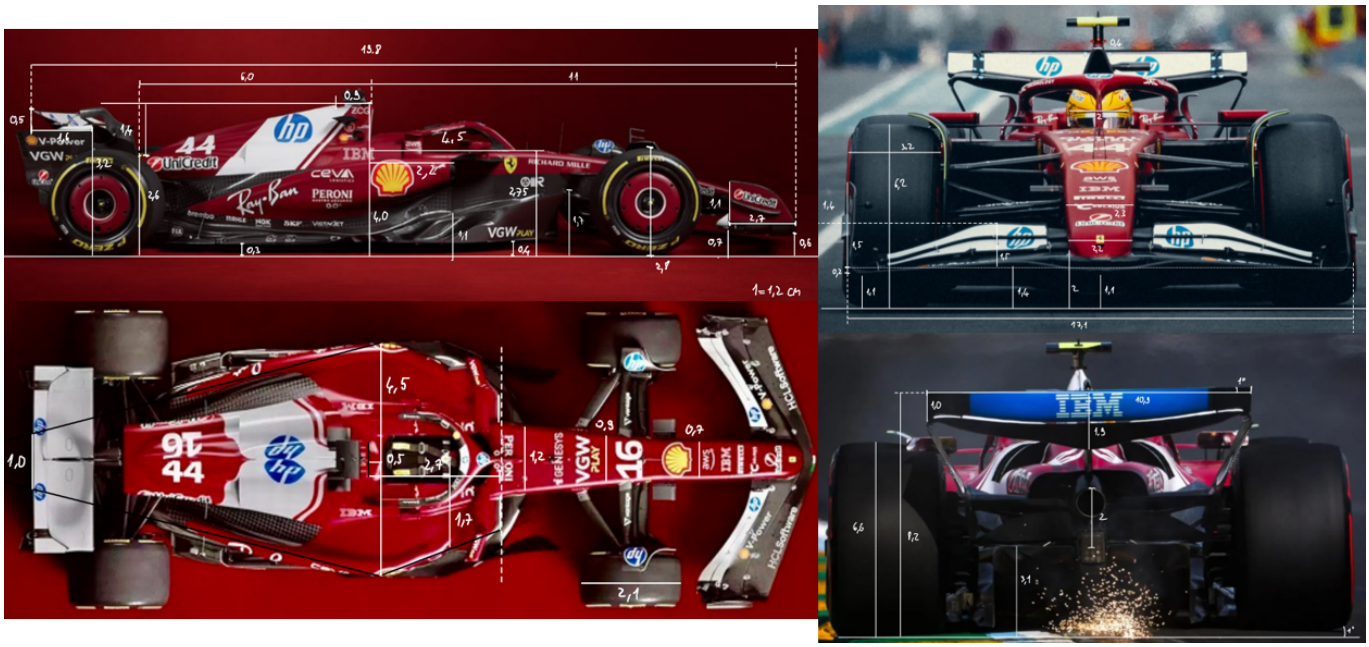


Figure 3: Car measures for design the 3D CAD (images sources: [7],[8], [9], [10])

All references are taken from the Ferrari SF25 [3]. To translate measures in meters, the radius and width of wheels have been used, since they are well defined and known. It is important to specify that there may be errors given by different photos scales, but since having a very precise model is not required for this study, they have been neglected. If it is complicated to find measures for the general car, wing characteristics are more difficult to obtain, since they change in function of the car and circuit. Therefore, the modeling of the front and rear wing was made approximately by pictures without any measure. As we can see from the top view of the car in figure 3, the front wing has a v-shape. But also in this case, since the focus is in the 2D profile, it is modeled straight. The references [4] are used only to determine the position of the four wing elements, their shape and orientation.

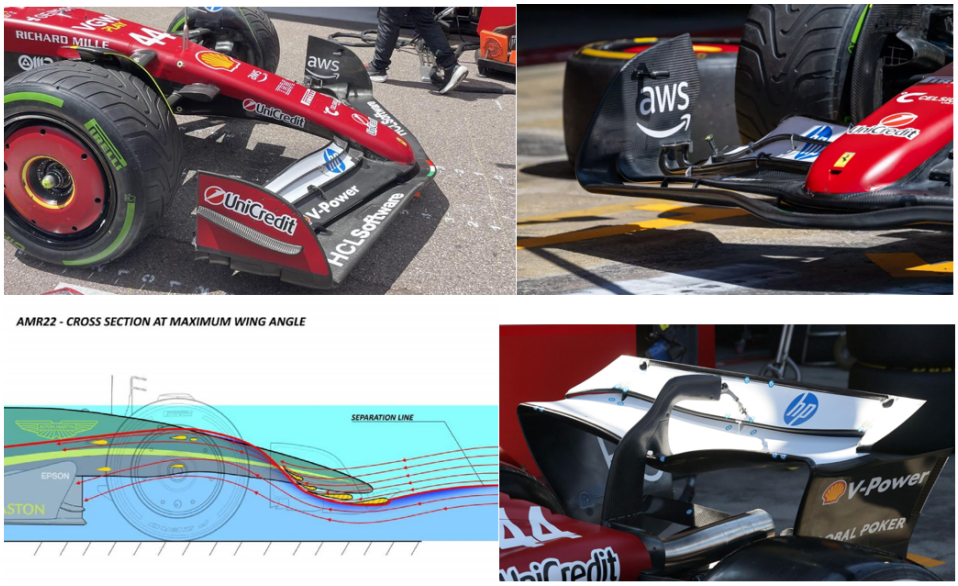


Figure 4: Wings references (images sources: [11],[12], [13], [14])

To model the profile of the wings, a Fusion 360 toolbox has been used to generate NACA airfoils: Airfoil Generator by Autodesk [15]. The front wing is composed of a NACA 2410, NACA 3710 and two NACA 4810, from the bottom to the top, respectively. A NACA 9510 and a NACA 6510 form the rear wing.

B. *Unit system*

The CAD model is designed in millimeters, but the reference scales chosen for this study are aligned with the international system (SI), namely meters for length, seconds for time, kilograms for mass and newton for force.

C. *Characteristic dimensions*

The characteristic length used for computations is the car length which is of 5 meters. The speed selected for this study is of 103.266 m/s (372 km/h), which is very close to the all time speed record during a Formula One race, achieved by Valtteri Bottas with the Williams Racing FW38 in 2016 at the Mexican Grand Prix [16]. This speed is not associated with 2025 cars, but represents the maximum capability of a Formula One car in terms of speed performance. Since the study purpose is to understand if the aerodynamics of the car is able to counteract gravity, the choice of this extreme velocity can compensate for large approximations on the car model, keeping feasible conditions. This velocity corresponds to a Reynolds number of $3.5 \cdot 10^7$, which suggests a turbulent environment. Another important length to consider is the distance from the ground. In this study, the bottom of the car is at a minimal distance of 70 mm and the front wing is placed at height of around 140 mm.

D. *Geometrical space*

As already specified previously, the geometrical space considered in this study is two dimensional.

III. Physical modeling and hypotheses

A. *Fluid model and physical assumptions*

The fluid is modeled as Newtonian, incompressible air, under the assumption of constant thermophysical properties. The dynamic viscosity is set to

$\mu = 1.789 \times 10^{-5}\,\mathrm{kg\,m^{-1}\,s^{-1}}$

and the density to

$\rho = 1.225\,\mathrm{kg\,m^{-3}}$

corresponding to standard atmospheric conditions at sea level.

Since the objective of the study is limited to the aerodynamic flow behavior around the vehicle, the governing equations are restricted to the incompressible Navier–Stokes equations. Thermal effects, heat transfer, and chemical reactions are neglected, as they do not significantly influence the flow physics relevant to lift and pressure force generation in the present configuration.

B. *Turbulence modeling*

The characteristic Reynolds number of the flow is sufficiently high over the entire computational domain to require an appropriate choice of turbulence modeling [17]. Several *Reynolds-Averaged Navier–Stokes (RANS)* models were therefore theoretically considered.

Although the one-equation *Spalart–Allmaras* [18] model is commonly employed in external aerodynamic applications, despite its robustness and low computational cost, it was deemed insufficient for accurately capturing the complex flow behavior near the car surface. In particular, the analysis focuses on boundary layer region phenomena, near-wall pressure gradients, and the resulting aerodynamic forces, which require improved near-wall resolution. Moreover, flow separation is expected, as well as recirculation in certain areas.

For this reason, the two-equations $k-\omega$ *Shear Stress Transport (SST)* model was selected [19]. This model combines the advantages of the Realizable $k-\epsilon$ formulation [20] in the free-stream region with the superior near-wall accuracy of the standard $k-\omega$ model. In particular, the $k-\omega$ SST model does not rely on wall functions, allowing for a more accurate treatment of boundary layer effects when appropriate mesh refinement is applied [21].

IV. Boundary conditions, external forces, initial conditions

A. *Boundary conditions*

In this study, the car is considered to run at constant speed of 103.266 m/s (372 km/h). Hence, to include the movement of the car into the simulation, the car is considered at rest and the fluid travels from the right to the left at car speed. Therefore, the boundary condition at the inlet (the right side of figure 1a) is set as inward velocity at 103.266 m/s. Since at the outlet the velocity is not known and considering that it is located far away from the car, the outlet absolute pressure is set to be at ambient pressure of $1.01 \cdot 10^5$ Pa. The top side of the fluid region has a symmetry boundary condition. The whole car and the floor have a no slip boundary condition, but since the car is not at rest, the ground is set to move at the car speed, hence a velocity of 103.266 m/s directed from right to left was selected.

B. *External forces*

The only force acting on the system is gravity. However, considering that it acts on air and the high speed, its effect is neglected. Therefore, there are no external forces acting on the fluid in the simulation.

V. Computational mesh

A. *Mesh and cell types*

To address the complex geometry of certain car components, such as the wings, an unstructured mesh consisting mostly of triangular elements was employed. The only structured quadrilateral elements appear in the boundary layer, resulting in an hybrid mesh. This choice also ensured precise control over local refinement, which is essential in applications where the computational domain is significantly larger than the geometry of the car itself.

B. *Size and number of cells*

The starting point of this study was a mesh generated with global sizing parameters, with maximum and minimum element sizes set to $500\,\mathrm{mm}$ and $250\,\mathrm{mm}$, respectively. Consistent with expectations, this preliminary simulation failed to yield physically realistic results, compounded by poor skewness and orthogonality metrics. Then, several local sizing controls were defined, focusing on the most geometrically complex parts of the model and on the regions most influential to the objectives of our study. In Table I, the target size parameter for each of the local sizing implemented in our final mesh are reported. These values were initially chosen based on the dimensions of the surrounding elements. For example, in the two wing regions, the target size was set to one fifth of the length of the smallest wing. After further simulations, the target sizes of the regions exhibiting the highest velocity magnitudes and gradients were reduced, as well as in the areas most relevant to our main research question, such as the space between the car and the ceiling. A body of influence was also introduced to separate the zone closest to the car from the rest of the domain.

Zone Label	**Target Size** [mm]
Front Wings	5
Rear Wings	10
Ceiling Area	50
Body of Influence	100
Car Back, Car Front, Car Under Spoiler, Recirculation Area	25
Ceiling Downstream, Ceiling Upstream	250
Ceiling Under Car	25

Table I: Adopted local sizings, with the respective target size parameter chosen. Figure 5 helps clarify the meaning of the labels

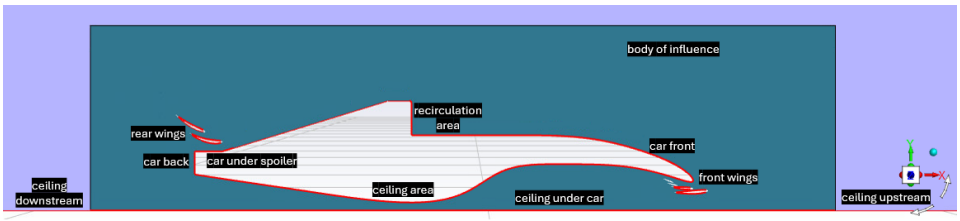


Figure 5: Illustration of the zone labels

Finally, to more accurately capture the complex flow phenomena near the wall boundaries, a uniform boundary layer was added on both the ceiling and the car edges. The boundary layer parameters were obtained using the online calculator available on the Fluid Mechanics 101 website [22]. This computation yielded the values illustrated in Table II.

Parameter	Computed Value
Number of Layers	20
Growth Rate	1.4
First Layer thickness	0.00872 mm
Maximum Layer Height	19.7 mm

Table II: Boundary layer parameters, computed via Fluid Mechanichs 101 website

As a result, our final mesh contains 33274 cells and 26906 nodes.

C. *Final Mesh*

Figure 6 illustrates the overall mesh configuration, while Figure 7 shows a close-up of the mesh within the body of influence. A visual assessment of the mesh indicates a regular and well-refined discretization in the regions of interest.

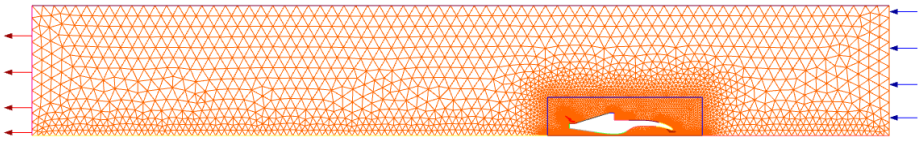


Figure 6: Visualization of the final mesh

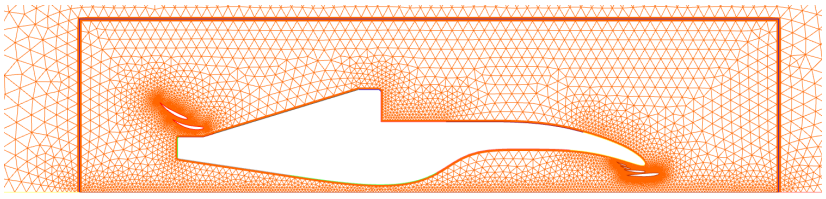


Figure 7: Visualization of the final mesh in the vicinity of the car

In Figures 8, 9 and 10, the histograms of three mesh quality metrics are presented: orthogonality, skewness, and aspect ratio. Table III reports the corresponding average and worst values for each metric. It can be observed that the average values of orthogonality and skewness are very good: orthogonality is above the optimal threshold of 0.95, while skewness remains comfortably below the threshold of 0.25. The same does not hold for the worst values, which fall outside the acceptable ranges (orthogonality < 0.05 and skewness > 0.85). However, as can be seen from the histograms, the number of elements with critical values is very small and confined to specific regions, where the car geometry is highly complex and detailed at a small scale. In particular, this occurs in the vicinity of the wings, as illustrated in Figure 11. This issue may be mitigated by further mesh refinement in these regions; nevertheless, it cannot be fully eliminated, as it is inherent to the geometry and cannot be modified without adversely altering it. Therefore, refining the mesh would have required significantly increasing the number of cells for a negligible improvement in the quality metrics. For these reasons, it was decided not to further modify the mesh and proceed while accepting the presence of these few low-quality elements. Regarding the aspect ratio, both the average and worst values are quite high. However, this is largely a consequence of the very thin cells in the boundary-layer region, whereas far from the walls the aspect ratios are comfortably within the acceptable range. This behavior is unavoidable when modeling such boundary layers, where the first cell must be very thin in order to achieve acceptable values of the y^+ parameter. Since accurately

capturing near-wall behavior is crucial, and these thin, high-aspect-ratio elements are consistent with the physics—characterized by very large gradients in only one direction—these high aspect ratio values near the wall can be considered acceptable.

Quality Metric	Average Value	Worst Value
Orthogonality	0.9562	0.001311
Skewness	0.06019	0.8759
Aspect Ratio	381.480	11943.9

Table III: Average and worst values of mesh quality metrics

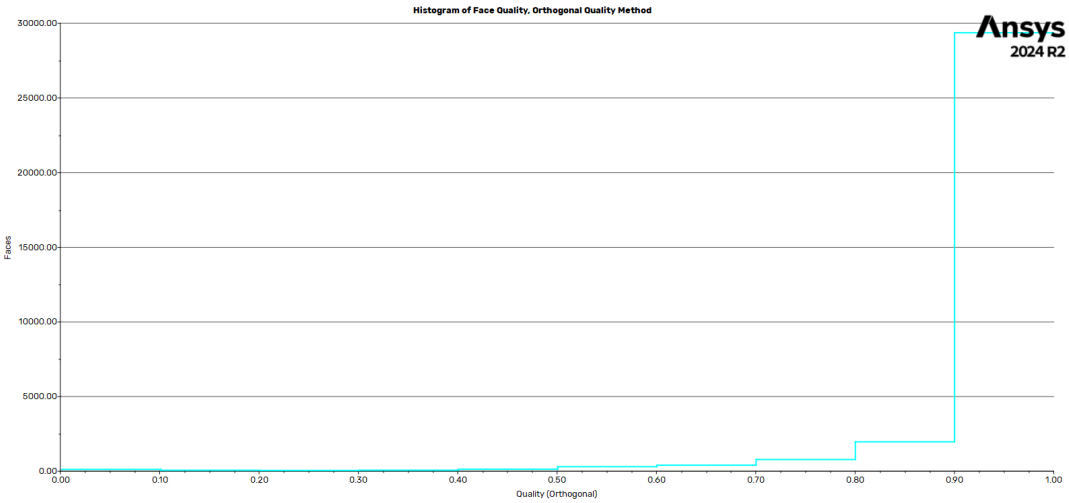


Figure 8: Histogram of the orthogonality of the obtained cells

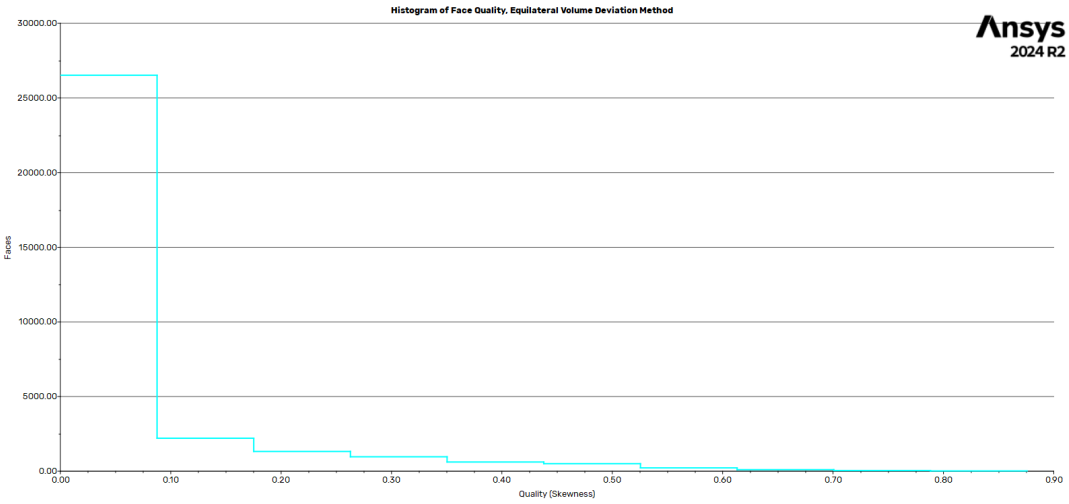


Figure 9: Histogram of the skewness of the obtained cells

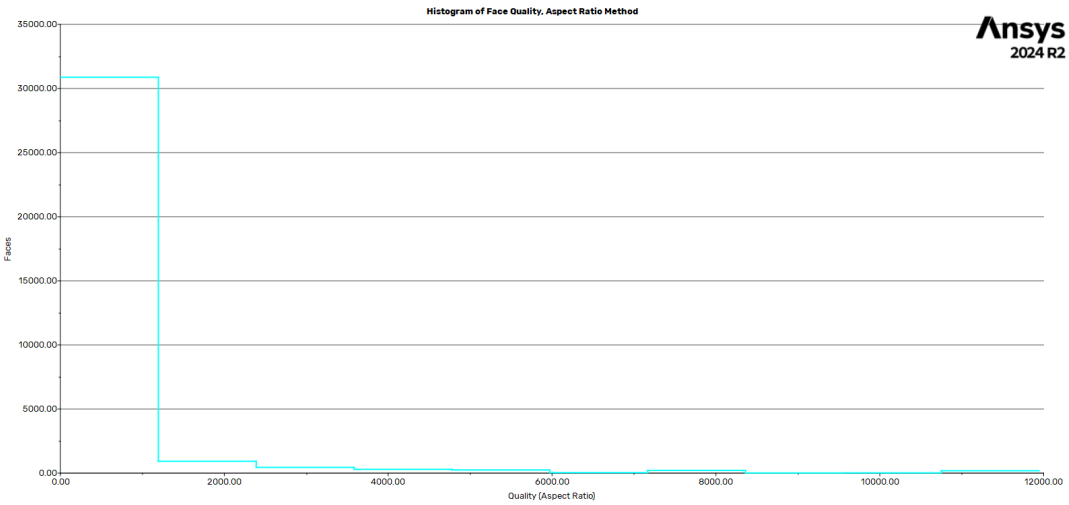


Figure 10: Histogram of the aspect ratios of the obtained cells

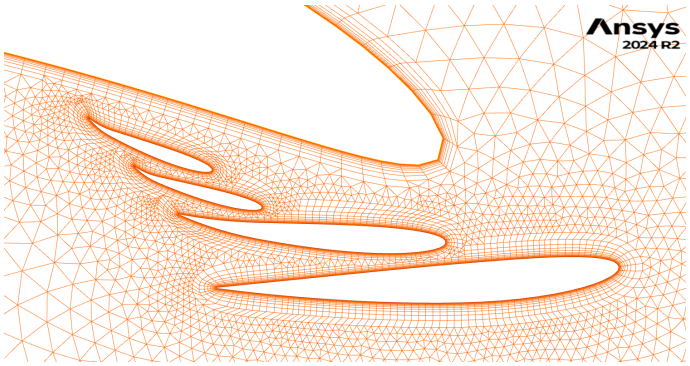


Figure 11: Detail of the mesh in the vicinity of the front wings

VI. Numerical methods

A. *Spatial discretization methods*

The Finite Volume Method was adopted to discretize and solve the Navier-Stokes equations over the computational mesh, using ANSYS Fluent as computational tool. To compute gradients appearing in the diffusive terms, the Least Squares Cell-Based method was employed, as it provides higher accuracy on the unstructured mesh adopted in this work. Moreover, the convective terms in the momentum equation, turbulent kinetic energy k equation, and specific dissipation rate ω equation were discretized using a Second Order Upwind scheme on the final mesh (the k–ω SST turbulence model was used, as discussed later). The logic behind this approach is the following: a second-order discretization was selected after obtaining a mesh on which the solution converged with first order methods, to improve accuracy and reduce numerical diffusion, while maintaining stability thanks to the good quality of the mesh. [23] Following analogous reasoning, a Second Order discretization scheme was also chosen for the pressure gradient, allowing convergence towards a velocity-pressure field $(\mathbf{u}, p)$ that better satisfies the continuity equation [23], even though the coupled solver (discussed later) already provides a strong velocity-pressure coupling and physically consistent solutions. These choices result in an overall second-order accurate spatial discretization, except in a limited number of low-quality mesh cells, as shown in previous sections.

B. *Solver type and reference values*

Since steady, incompressible turbulent flow conditions were assumed, as described in Sections I and II, a steady, pressure-based solver in ANSYS Fluent was employed. To solve the coupled continuity–momentum equations, a coupled scheme was selected, in order to improve the physical consistency of the solution, avoid oscillations, and increase the convergence rate in steady-state simulations, providing a favorable trade-off between accuracy and computational cost with respect to a sequential update [24]. As explained in Section III.B, the k–ω SST turbulence model was adopted, using the default set of model constants. Accordingly, the variables solved for are the mean velocity field $\mathbf{u}$, pressure p, turbulent kinetic energy k, and specific dissipation rate ω. For clarity and reproducibility, we provide in the appendix the k–ω SST equations solved (default in ANSYS Fluent) and the default model constants used.
We chose as reference values the following:

- **Length**: the reference length for the computations was set as the length of the car, approximating it at $5\,\mathrm{m}$. (Note that the in Fluent this reference value is defined as *Area* for the 3D case).
- **Density**: the standard air density value [25] was chosen, $\rho = 1.225\,\mathrm{Kg\,m^{-3}}$.
- **Temperature**: A reference temperature of $T = 288.16$ K was adopted, corresponding to standard atmospheric conditions at sea level [25].
- **Velocity**: A reference value of $V = 103.266\,\mathrm{m/s}$ was chosen, considering the Formula 1 cars speed record [16].
- **Dynamic viscosity**: A reference viscosity of $\mu = 1.7894 \cdot 10^{-5}\ \mathrm{kg/(m\,s)}$ was adopted, corresponding to standard atmospheric conditions at sea level [25].

C. *Solution options*

A hybrid initialization of the solution was performed, since it is considered best practice when working with an unstructured mesh and a coupled solver in order to avoid pathological initial fields. Moreover, it provides an initialization of k and ω consistent with local velocity and length scales, preventing unstable behavior in the near-wall region [26] [27]. Indeed, our experiments show that it has been a valid choice. Normalized residuals were adopted as the primary convergence criterion. In particular, a tolerance of 10^{-3} was selected, which has proven to provide a sufficient level of numerical accuracy for the present steady-state simulations. We also imposed a maximum number of iterations equal to 1000 as secondary convergence criterion to avoid very long simulations.

D. *Computed quantities*

The following quantities were computed to evaluate the results obtained in our simulations, check mesh convergence and investigate our research question:

- **Pressure and Velocity Contours**: To check qualitatively the physical consistency of our steady solution, we computed pressure and velocity magnitudes along the entire domain.
- **Lift Coefficient**: To evaluate whether the car can drive on the ceiling under physically acceptable velocities and conditions, it is necessary to assess whether the aerodynamic downforce can overcome gravity. Therefore, the total lift coefficient acting on the whole car was evaluated and decomposed into pressure and viscous contributions. We calculated it with respect to $(0, 1, 0)$ direction in our domain.
- **Moment and Drag Coefficient**: The total moment and drag coefficients acting on the car, decomposed into pressure and viscous contributions, were also computed to assess mesh convergence and

to provide additional physically relevant quantities that complement the lift results. The coefficients were calculated with respect to $(0, 1, 0)$ and $(0, 0, 1)$ direction respectively.

- y^+: This quantity was computed in all the near-wall regions to further evaluate the quality of the mesh and the modeled solution consistency near borders.
- **Mass Flow Rate**: To further validate the physics of our results, the net mass flow rate was computed considering the flow across our entire domain.

VII. Mesh convergence study

A. *Criterion*

For the convergence study, three quantities relevant to our analysis were taken into account, so as to avoid the risk that their potentially different rates of convergence could lead to an incorrect assessment of the overall convergence. In particular, the following variables:

- Lift Coefficient
- Drag Coefficient
- Moment Coefficient with respect to the car's Center of Mass

B. *Presentation of the different meshes*

Five different meshes were considered. Starting from the coarsest one, the remaining meshes were generated by successively halving the maximum and minimum global mesh sizes, as well as the target sizes used for the local refinements. On the other hand, the boundary layer mesh was left unchanged across the refinements, since its parameters were specifically tuned for this problem and modifying them could have compromised the accurate capture of the near-wall flow behavior. In Table IV the sizes selected for the coarsest mesh are reported. As mentioned above, the sizes of the remaining meshes can be obtained by halving these values at each refinement step. The boundary layer parameters are the same as Table II for all the five meshes.

Global Sizing Parameters	**Value** [mm]
Maximum Size	4000
Minimum Size	2000
Local Sizing Zone	**Target Size** [mm]
Front Wings	40
Rear Wings	80
Ceiling Area	200
Body of Influence	800
Car Back, Car Front, Car Under Spoiler, Recirculation Area	200
Ceiling Downstream, Ceiling Upstream	2000
Ceiling Under Car	200

Table IV: Sizing parameters of the coarsest mesh

By way of example, in Figures 12 and 13, details of the coarsest and finest mesh, respectively, near the car are provided. Finally, Table V contains the number of cells and nodes obtained for each mesh.

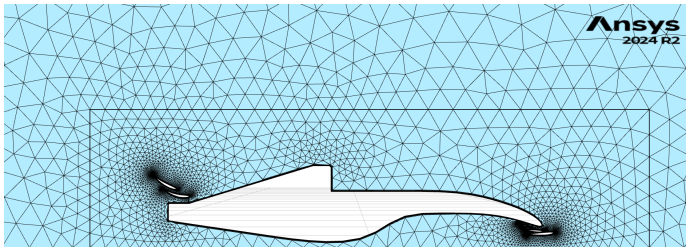


Figure 12: Visualization of the coarsest mesh in the vicinity of the car

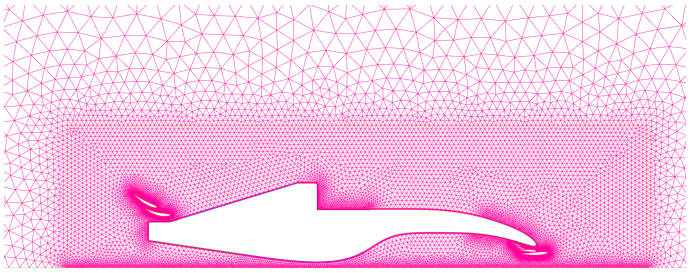


Figure 13: Visualization of the finest mesh in the vicinity of the car

Number of Refinements	Number of Cells	Number of Nodes
0	15296	12217
1	18109	14896
2	21496	17728
3	33274	26906
4	58133	42908

Table V: Number of cells and nodes of each mesh

C. *Results on the coarsest mesh*

Table VI reports the values of the three quantities of interest considered for the mesh convergence study on the coarsest mesh. Figures 14 and 15 represent contours of the velocity magnitude and pressure fields, respectively.

Quantity of Interest	Obtained Value
Lift Coefficient	-1.297
Drag Coefficient	-0.2309
Moment Coefficient	0.5447

Table VI: Values of the quantities of interest obtained with the coarsest mesh

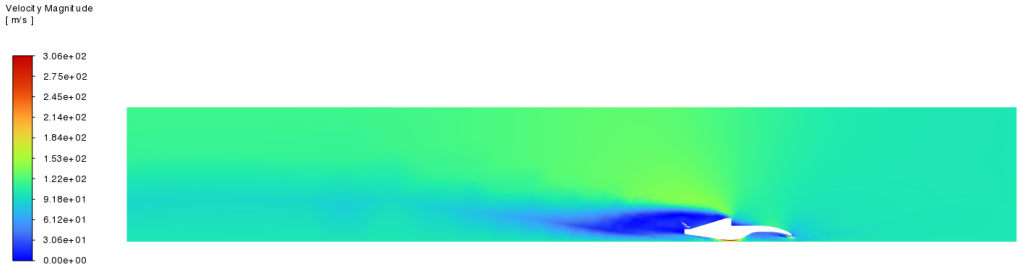


Figure 14: Contour of the velocity magnitude obtained with the coarsest mesh



Figure 15: Contour of the static pressure obtained with the coarsest mesh

An examination of the velocity-magnitude contour reveals several shortcomings. First, the insufficient mesh refinement results in pronounced numerical diffusion, which artificially smooths the velocity gradients and prevents a sharp representation of the flow field. Second, the contours appear patchy due to discretization artifacts. Finally, when compared with the contours obtained using finer meshes (see, for example, Figure 16), the discrepancies in the wake region are substantial, indicating that this area is poorly resolved in the present simulation.

D. Results on the finest mesh

Table VII reports the values of the three quantities of interest considered for the mesh convergence study on the finest mesh. Figures 16 and 17 represent contours of the velocity magnitude and pressure fields, respectively.

Quantity of Interest	**Obtained Value**
Lift Coefficient	-1.200
Drag Coefficient	-0.1994
Moment Coefficient	0.4002

Table VII: Values of the quantities of interest obtained with the finest mesh

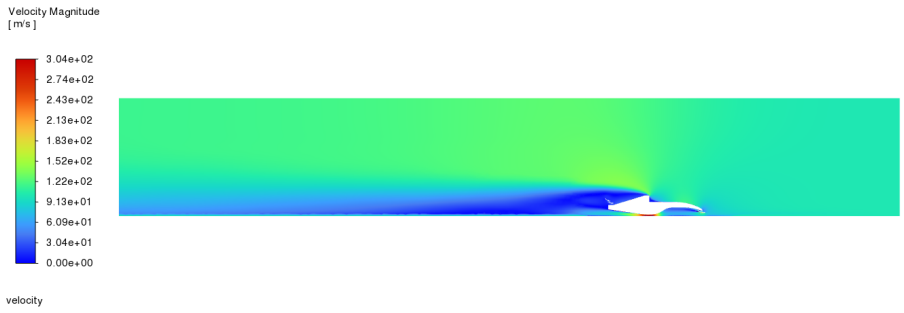


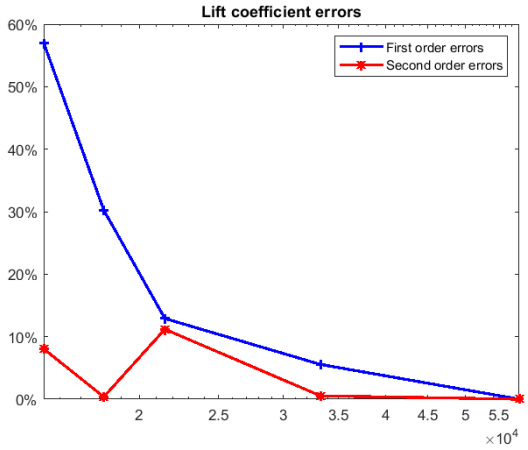
Figure 16: Contour of the velocity magnitude obtained with the finest mesh



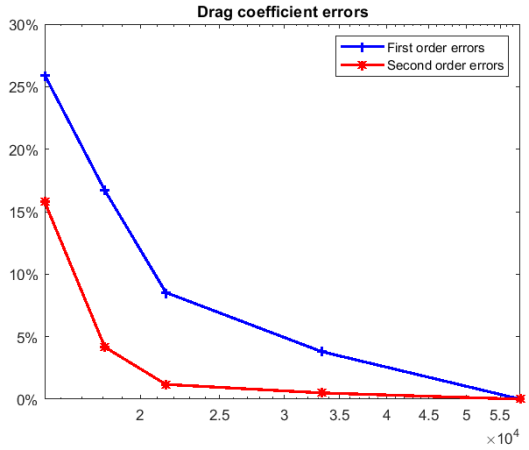
Figure 17: Contour of the static pressure obtained with the finest mesh

E. Mesh Errors and Variations

First, simulations on the five meshes were performed using standard spatial discretization for the pressure and first-order upwind discretization for all other variables. Then, the simulations were repeated using the default options, namely second-order discretization for the pressure and second-order upwind for the remaining variables. Note that all the results mentioned in Subsections C and D were obtained with the second order configuration. In Figures 18 and 19, the errors computed with respect to the finest mesh are reported, together with the incremental variations between successive meshes. These results will be used to guide the selection of the final mesh.



(a) Error estimates of lift coefficients with respect to the finest mesh

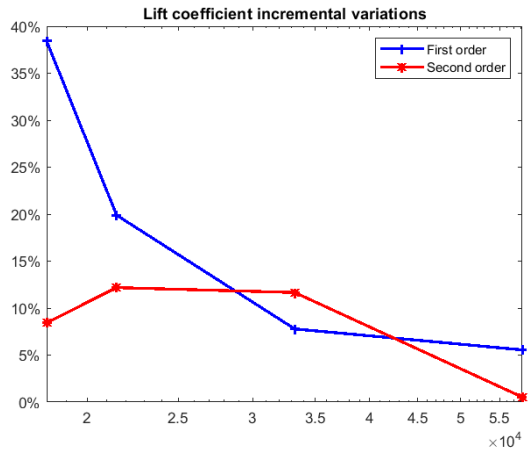


(b) Error estimates of drag coefficients with respect to the finest mesh

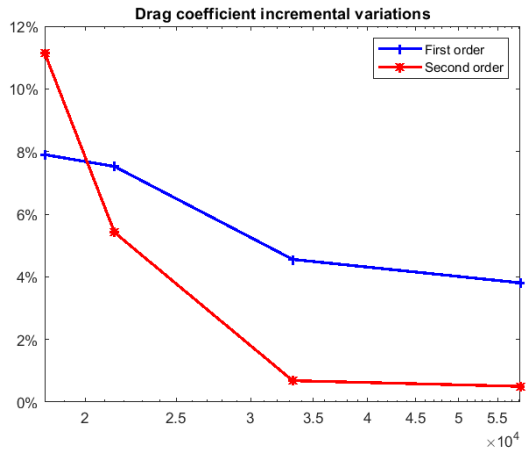


(c) Error estimates of moment coefficients with respect to the finest mesh

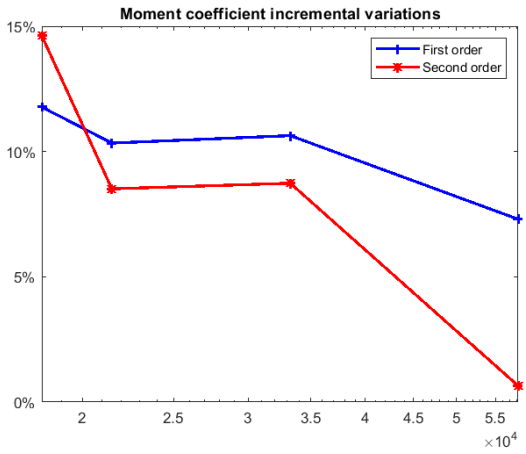
Figure 18: Plots of the error estimates of the quantities of interest with respect to the finer mesh. The x-axis represents the number of cells on a logarithmic scale.



(a) Incremental variation of lift coefficient from one mesh to the next



(b) Incremental variation of drag coefficient from one mesh to the next



(c) Incremental variation of moment coefficient from one mesh to the next

Figure 19: Plots of the incremental variations of the quantities of interest from one mesh to the next. The x-axis represents the number of cells on a logarithmic scale.

F. Choice of the mesh

As expected, the second-order configuration exhibited faster convergence with respect to the number of cells. Therefore, it was decided to rely exclusively on this configuration to assess convergence. A threshold of 3% (a value often used in industrial applications) was set for the error estimates and selected the mesh for which the errors associated with all three quantities of interest fall below this threshold. Following this procedure, the fourth mesh (i.e., the mesh obtained after three refinements of the coarsest grid) was selected. It was then verified that the relative differences between this mesh and the finest one were acceptable. Since all three values are comfortably below 2%, the selected mesh was adopted for the subsequent analyses.

VIII. Results

With the chosen mesh, the physical consistency of the solution was checked and tried to answer our research questions, computing physically meaningful quantities as described in section VI. Our results are reported below.

- **Pressure and velocity contours**: The velocity and pressure fields shown in Fig. 20 and Fig. 21

qualitatively indicate a physically consistent flow solution. The presence of ground effect [28] is clearly observed: the pressure reaches its global minimum beneath the vehicle, decreasing by nearly one order of magnitude precisely in the region where the flow accelerates to its maximum velocity. Downstream of the car, a well-defined and extended low-velocity wake develops, consistent with boundary-layer separation in the near-wake region.

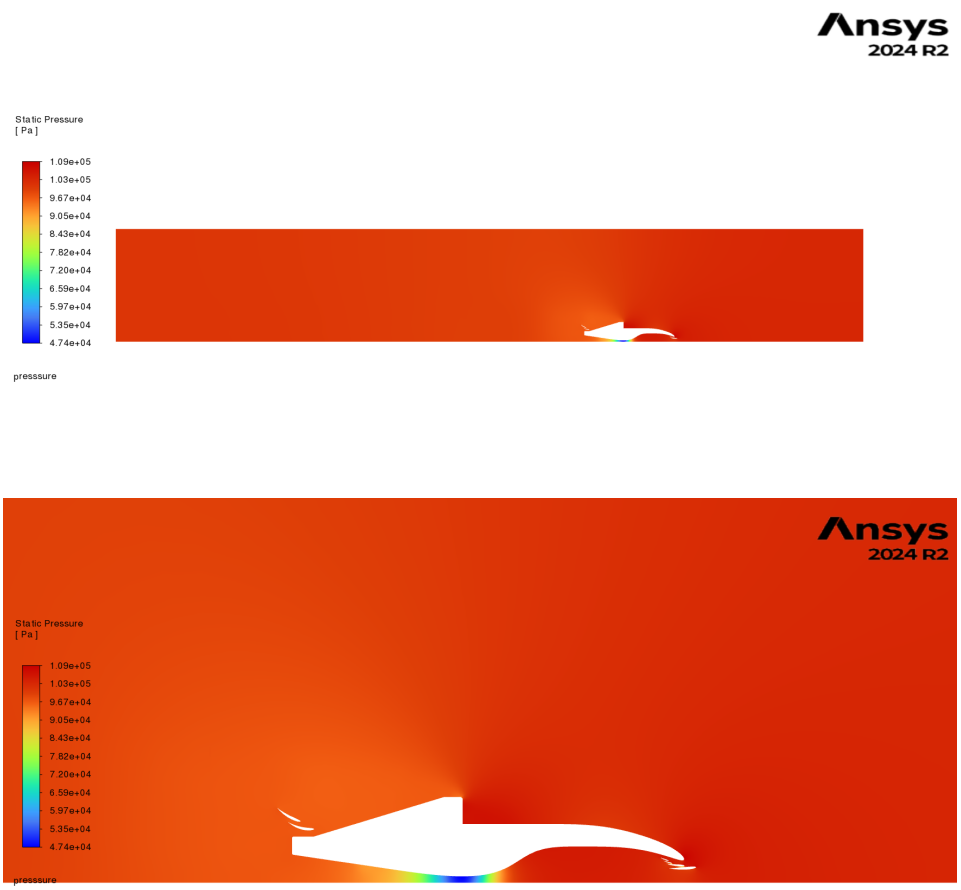


Figure 20: Pressure contour-final mesh.

- **Lift Coefficient**: In Tab. VIII the values obtained for the lift coefficient in the chosen mesh are reported, divided in pressure and viscous contributions, acting on the whole car and computed with respect to $(0, 1, 0)$ direction. It can be observed that the lift is almost entirely pressure-driven as expected, consistently with this configuration. The net lift coefficient is negative, showing that there is a downforce acting on the car towards the ceiling.

C_L **Contribution**	**Total Value on The Car**
Pressure	-1.2003154
Viscous	6.5921213e-05
Total	-1.2002495

Table VIII: Lift Coefficient in direction $(0, 1, 0)$ Acting on the Car.

- **Moment and Drag Coefficient** The moment and drag coefficients computed on the entire car using the selected mesh are reported in Tables X and IX, respectively, with respect to the $(0, 0, 1)$ and

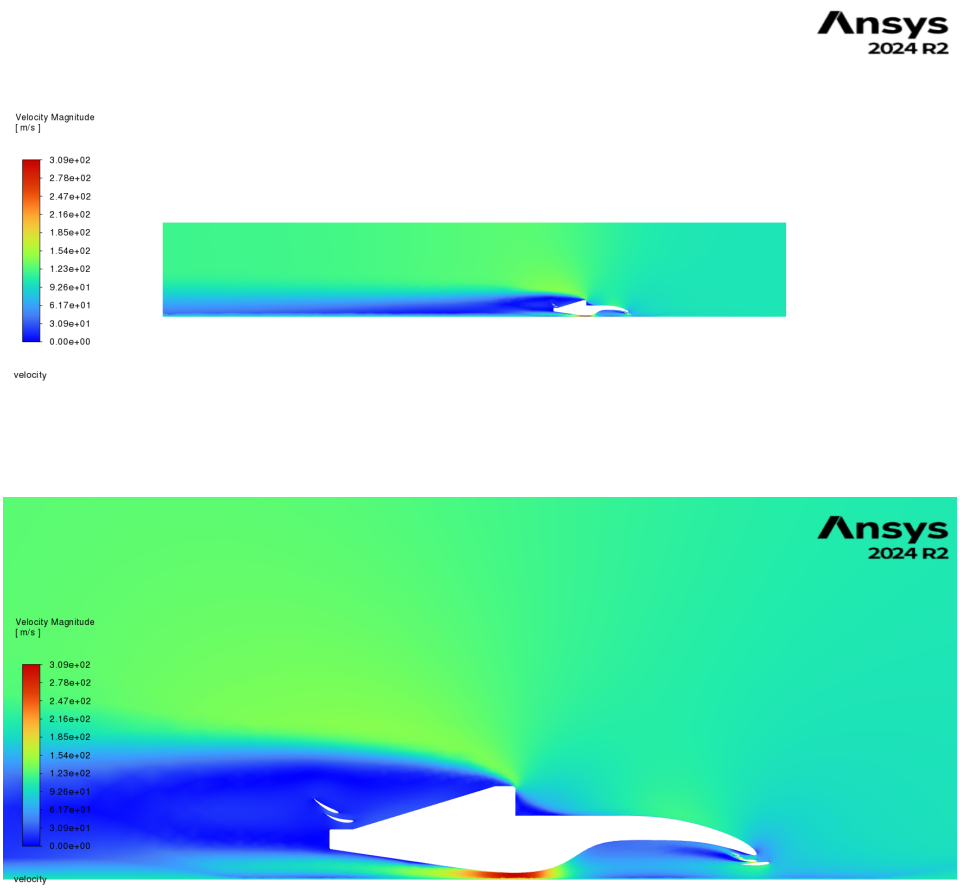


Figure 21: Velocity contour-final mesh.

$(1, 0, 0)$ directions. As expected for the present setup, pressure represents the dominant contribution. The total drag coefficient is negative, indicating a drag force acting opposite to the direction of motion of the car.

C_D **Contribution**	**Total Value on The Car**
Pressure	-0.19238888
Viscous	-0.0059389802
Total	-0.19832786

Table IX: Drag Coefficient in direction $(1, 0, 0)$ Acting on the Car.

C_M **Contribution**	**Total Value on The Car**
Pressure	0.40129008
Viscous	0.0014901922
Total	0.40278027

Table X: Moment Coefficient Acting on the Car.

- y^+: Figure 22 shows the near-wall y^+ values for each zone. A near-wall approach based on the SST turbulence model was adopted to model turbulent flow behavior close to walls and boundaries. The figure shows that $y^+ = \mathcal{O}(1)$, indicating that the generated mesh is well designed and that the

obtained values are consistent with the requirements of the SST near-wall treatment. In particular, the highest y^+ values are observed in the zones where few lower-quality mesh cells are present, so in the front and rear wings regions.

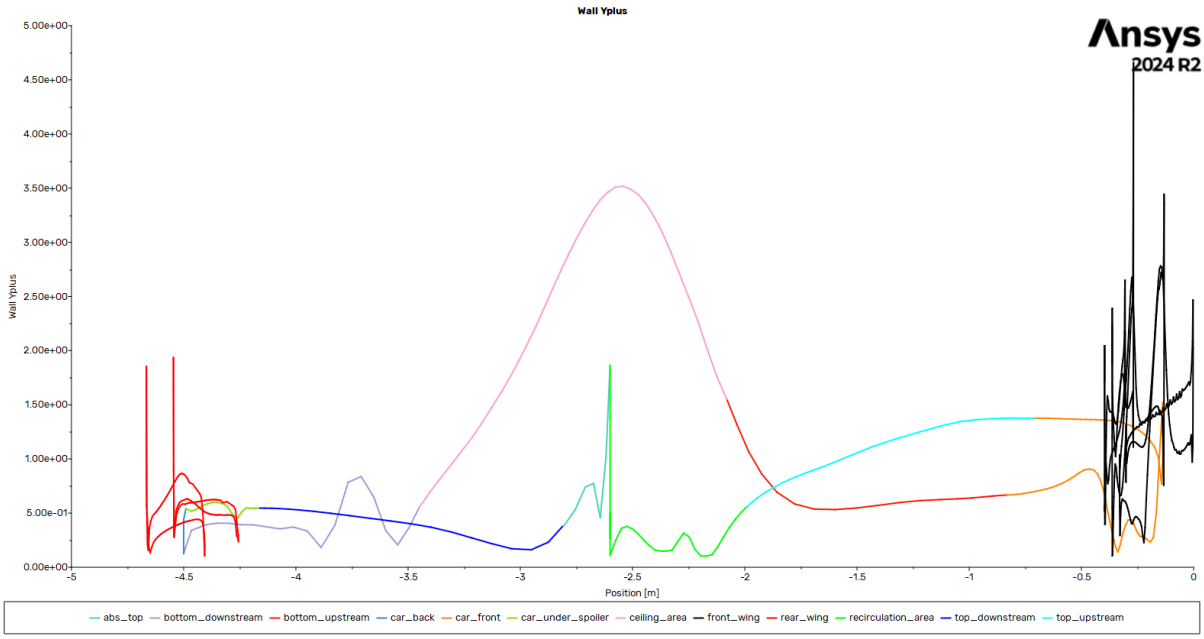


Figure 22: Near-the-wall y^+ values per zone.

- **Mass Flow Rate**: To further investigate the physical consistency of our simulation, the net mass flow rate across the whole domain was calculated. The values at inlet, outlet and the net are listed in Tab. XI (the flow across other boundaries is obviously zero). We obtained a net mass flow rate ≈ 0, additionally validating our results.

Boundary	Mass Flow Rate
Inlet	689.77415
Outlet	-689.86585
Net	-0.091703164

Table XI: Mass Flow Rate Across Domain.

IX. Analysis and Conclusions

A. *Summary of calculated results*

The main purpose of this study was to simulate the flow around a Formula One car in order to understand if the aerodynamic performance allows the car to run upside down against gravity. Therefore, the result of interest is the lift coefficient which is of -1.200. Other quantities considered more for convergence reasons than physical, are the drag coefficient and the moment coefficient, which values are -0.198 and 0.403 respectively.

B. *Criteria and analysis*

Formula One teams do not share their data to the public for competitive reasons, therefore it is not possible to compare the result found in this study with some real values. However, different CFD studies are available. Despite studies involving the whole Formula One car 2D-profile are unavailable, some

two-dimensional studies discussing only some element of the car can be used as reference. For example *Study on airflow characteristics of rear wing of F1 car* by Azmi et al provides two-dimensional results related to the rear wing [29]. Results and experimental values of the lift coefficient provided in the study are around -0.20. Considering that this value corresponds to the rear wing only in a free stream, it can be stated that a lift coefficient of -1.20 for the whole car is a comparable result. Same kind of comparison can be made with another study published by Granados-Ortiz et al, where a two-dimensional analysis on a 5-element F1 car front wing is analyzed [30]. In this study, lift coefficient values for different configurations of the front wing are between -0.112 and -0.372. Even in this case, considering that these values are obtained from the front wing only, the result of -1.20 for the whole car is reasonable.
Figure 20 shows that the main lift source in this simulation is given by the ground effect under the car. In fact, a huge pressure drop can be visualized, which pushes the car against the floor. Taking into consideration the results just mentioned, the lift contribution given by the ground effect, considering the extreme case, should be of the order of $1.20 - 0.20 - 0.37 = 0.63$, resulting in 52.5% of the total lift. This qualitative value can be considered an underestimation of the effective ground effect contribution, since by looking at Figure 21, in the rear wing area, the flow is very slow compared to the rest suggesting an high turbulent environment, that will lead to a much lower lift contribution. Therefore, it can be stated that this study result is valid.
Studies related to the whole car can be found but in a three-dimensional environment. The thesis of Unmukt Rajeev Bhatnagar published by the Pennsylvania State University, shows three-dimensional lift coefficient values for different front and rear height [31]. The result of 1.20 for the lift coefficient is significantly lower compared to the minimal value of the reference of 2.16. This huge difference can be attributed to the dimension of the study: Bhatnagar study is placed in a three dimensional environment, whereas this study focuses on a two dimensional profile. The two dimensional study therefore seems to be limiting. All transversal contributions are neglected. For example, as already specified, in this simulation the rear wing does not give an important contribution in generating lift. Figure 23 shows pathlines around the car. An interesting behavior can be observed in the upper back area of the car. The fluid coming from the front is lifted by the top corner of the car and given the high speed, it does not follow the car back. Therefore, it leaves space in that area generating low pressure. This mechanism allows the fluid coming from the back of the car to climb from the back, under the rear wing and then fill the void left by the fluid running on top of the car. This phenomenon will be limited in a three dimensional simulation, since the transversal fluid coming from the sides of the top part of the car will better follow the car shape and then pass between the rear wing components generating an additional lift contribution.
But it is important to specify that the comparison between two- and three-dimensional lift coefficients does not give relevant information for the validation of the result. It gives an approximation of the range of values and helps to understand if the result is admissible.

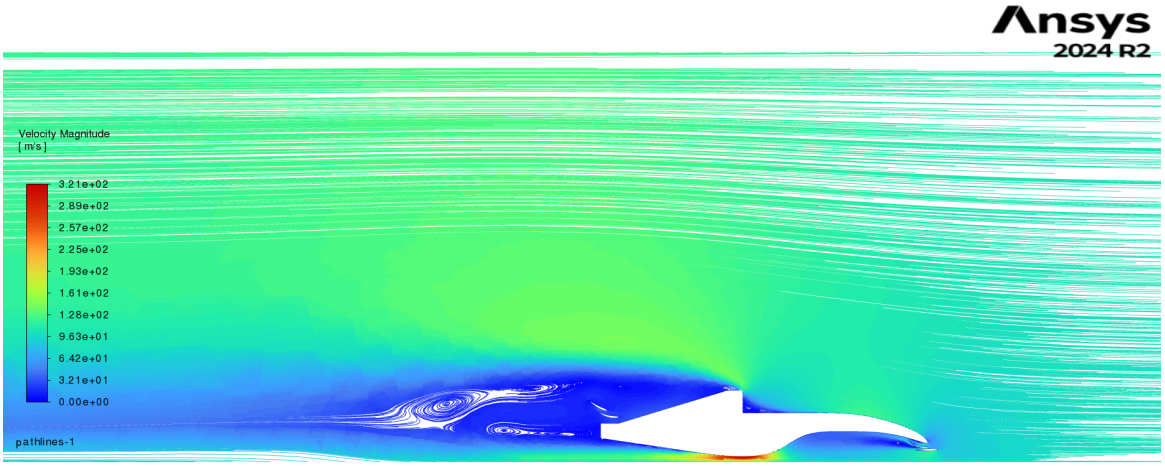


Figure 23: Pathlines of the fluid around the car

More interesting can be the comparison of the total lift generated on the car. For a three-dimensional analysis, the lift is calculated by

$$L_{3D} = \frac{1}{2}\rho V^2 C_L S \tag{1}$$

whereas the two-dimensional lift is computed as

$$L_{2D} = \frac{1}{2}\rho V^2 C_l c \tag{2}$$

These results are still not comparable, since the three-dimensional lift refers to the whole body, whereas the 2D-lift acts on a very thin profile. In order to compare them, a very large approximation of the three-dimensional lift starting from the two-dimensional lift can be attempted. As just mentioned, the two-dimensional lift refers to a very thin slice of the whole object. Therefore, a large approximation of the three-dimensional lift can be made by multiplying the 2D-lift with the width of the object. This will certainly lead to an overestimation and is only useful for comparison purposes. A reduction factor δ should be added taking into account the variation of the geometry along the width and the three-dimensional effects such as vortex and transversal flow motion, which reduce the lift compared to the only extrusion of the profile.

$$L'_{3D} = L_{2D} \cdot w \cdot \delta \tag{3}$$

This method is applied with the simulations conditions specified in section VI, namely a characteristic length $c = 5$ m, air density $\rho = 1.225\, kg\, m^{-3}$ and velocity $V = 103.266$ m/s and adding the car width $w = 1.9$ m. For Bhatnagar thesis, the minimum lift coefficient value is considered. The study is referred to a 2010 F1 car, which is slightly smaller compared to the 2025 model, but these differences are neglected and the 2025 model measures are applied. These assumptions will lead to an underestimation of the lift.

$$L_{3D} = \frac{1}{2} \cdot 1.225 \cdot 103.266^2 \cdot 2.16 \cdot (2 \cdot 5) = 141'082.96\, N \tag{4}$$

The two-dimensional lift given by the simulation is

$$L_{2D} = \frac{1}{2} \cdot 1.225 \cdot 103.266^2 \cdot 1.20 \cdot 5 = 39'189.71\, N/m \tag{5}$$

And applying the approximation formula neglecting for the moment the reduction factor, the three-dimensional lift is

$$L'_{3D} = 39'189.71 \cdot 2 = 78'379.42\, N \tag{6}$$

The result obtained is 44.5% lower compared to the 3D-study, despite the absence of the reduction factor. This difference can be explained by the geometry of the model. A three-dimensional model takes into account the entire parts of the car that in a 2D-model are not present, for example the optimized shape of the front and rear wing or the wind channels on the floor.

To summarize, the result of this simulation compared to two- and three-dimensional studies can be considered valid for the purpose of this study. For a more precise and realistic simulations, other methods and models should be considered.

C. *Conclusions*

The main focus of this study is to understand if the aerodynamic performance allows a Formula One car to run upside down. For this phenomenon to be possible, the lift generated should be bigger compared to the gravitational force acting on the car. Article 4.1 of the 2025 FIA technical regulations state that the minimum mass is of 800 kg without fuel and driver [32]. Therefore, considering the driver and a small amount of fuel to run the car, the total mass of the car considered for the gravitational force is taken as 900 kg. Hence, the gravity acting on the car is

$$F_g = m \cdot g = 900 \cdot 9.81 = 8'829\, N \tag{7}$$

This result is much smaller compared to the three-dimensional lift of 78'379.42 N. Moreover, the gravity is smaller than the 2D-lift, which is 39'189.71 N/m. This is a different quantity, it represent the lift per unit length, whereas the 3D-lift is a force. But since the two-dimensional value is bigger than the gravity, it implies that it will be bigger than the gravity per unit length as well.
Therefore, it can be stated that a Formula One car has the capability to run upside down.

X. Discussion and Limitations

A. *Limitations*

The results of this study are inherently limited by the adopted modeling assumptions, most notably the use of a two-dimensional planar geometry. Such a simplification cannot fully capture the complexity of the flow field generated by a racing car. Turbulent flows are intrinsically three-dimensional, and important aerodynamic mechanisms—such as spanwise flow, vortex generation, and side-edge effects—are neglected in a 2D framework [33].

Furthermore, with specific reference to the motivating research question, a three-dimensional geometry would likely lead to significantly different results. The presence of lateral flow around the vehicle would alter the pressure distribution on the car surface and could either enhance or reduce the overall lift, depending on the interaction between underbody flow and side structures. As a result, the quantitative validity of the aerodynamic coefficients obtained in this study is limited.

Nevertheless, from a numerical standpoint, the simulations exhibit good internal consistency. The computed lift coefficients remain within a physically reasonable range when compared with values reported in more advanced numerical studies, suggesting that the adopted approach is suitable for qualitative analysis and methodological assessment [34].

B. *Future developments*

Several improvements could be implemented to enhance the accuracy and scope of the present study. A first step would be the extension of the model to a three-dimensional geometry, allowing the investigation of inherently 3D flow structures and their impact on aerodynamic forces. Such an approach would significantly improve physical realism, albeit at a higher computational cost.

To limit this increase in computational expense, the mesh could be further optimized by employing a hybrid strategy, with structured or quadrilateral elements in free-flow regions and refined unstructured meshes near the car surface. This would improve numerical efficiency while preserving solution accuracy in regions of interest.

Additionally, improved convergence and numerical stability for more complex configurations could be achieved by adopting problem-specific discretization schemes and solver settings, for example through a tailored implementation in OpenFOAM. Finally, for a more detailed analysis of turbulent flow phenomena, higher-fidelity and more computationally demanding approaches—such as more advanced RANS formulations or scale-resolving methods—could be considered (see, for example: [35], [36]).

Appendix

k–ω SST equations and model constants

We describe below the k–ω SST equations used in our simulations (default in ANSYS Fluent).

$$\begin{cases} \dfrac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho \mathbf{u} k) = P_k - \beta^* \rho k \omega + \nabla \cdot \left[(\mu + \sigma_k \mu_t) \nabla k \right], \\ \dfrac{\partial(\rho \omega)}{\partial t} + \nabla \cdot (\rho \mathbf{u} \omega) = \alpha \dfrac{\omega}{k} P_k - \beta \rho \omega^2 + \nabla \cdot \left[(\mu + \sigma_\omega \mu_t) \nabla \omega \right] + 2(1 - F_1) \rho \sigma_{\omega 2} \dfrac{1}{\omega} \nabla k \cdot \nabla \omega. \end{cases}$$

where:

- k — turbulent kinetic energy.
- ω — specific dissipation rate.
- P_k — production term of turbulent kinetic energy.
- α — model constant weighting the production of ω.
- β — model constant controlling the dissipation of ω.
- β^* — model constant controlling the dissipation of k.
- μ — dynamic viscosity.
- $\mu_t = \rho \frac{a_1 k}{\max(a_1 \omega, S F_2)}$ — turbulent (eddy) viscosity.
- σ_k — turbulent Prandtl number for k.
- σ_ω — turbulent Prandtl number for ω.
- $\sigma_{\omega 2}$ — turbulent Prandtl number for ω in the outer region (SST cross-diffusion).
- F_1 — blending function activating the k–ω formulation in the near-wall region.
- F_2 — blending function used in the limiter formulation of the turbulent viscosity.

We report in Table XII the default model constants used.

Constant	Symbol	Value	Role in the model
TKE dissipation constant	β^*	0.09	Controls the dissipation term $-\beta^*\rho k\omega$ in the k-equation.
Specific dissipation constant (inner)	β	0.075	Controls the dissipation term $-\beta\rho\omega^2$ in the ω-equation (near-wall region).
Specific dissipation constant (outer)	β	0.0828	Controls the dissipation of ω in the outer region (SST blending).
Production coefficient	α	0.52	Weights the production term $\alpha\frac{\omega}{k}P_k$ in the ω-equation.
Turbulent viscosity limiter	a_1	0.31	Limits the turbulent viscosity through $\mu_t = \rho\frac{a_1 k}{\max(a_1\omega, SF_2)}$.
Turbulent Prandtl number for k (inner)	σ_k	1.176	Controls the diffusive flux $(\mu+\sigma_k\mu_t)\nabla k$ near the wall.
Turbulent Prandtl number for k (outer)	σ_k	1.0	Controls the diffusive flux of k in the outer region.
Turbulent Prandtl number for ω (inner)	σ_ω	2.0	Controls the diffusive flux $(\mu+\sigma_\omega\mu_t)\nabla\omega$ near the wall.
Turbulent Prandtl number for ω (outer)	$\sigma_{\omega 2}$	1.168	Appears in the cross-diffusion term $2(1-F_1)\rho\sigma_{\omega 2}\frac{\nabla k\cdot\nabla\omega}{\omega}$.
Production limiter clip factor	–	10	Limits excessive production of k in high-strain regions to ensure numerical stability.

Table XII: Default model constants of the k–ω SST turbulence model and their role in the governing equations.

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